

Introduction to Exponential Random Graph Models

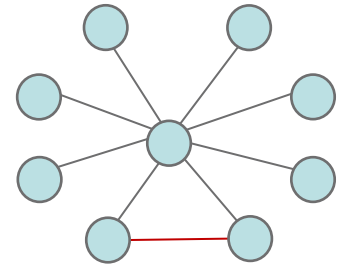
Networks and Innovation Training Programme
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Why ERG models?

Traditional models of tie formation do not take into account that **dyads are interdependent**.

- The presence or absence of a tie in one dyad may depend on the presence or absence of a tie in another dyad, for example because a single individual is involved in multiple dyads.



Traditional models cannot easily measure the effect of **structural mechanisms** of tie formation beyond the level of the dyad.

The basic idea

- The probability of observing a specific graph “x(obs)” is dependent on local characteristics of the graph ($f_i(x_obs)$)/

Effect of tie formation mechanism 1 in observed network

$$P(X = x_{obs}) = \frac{e^{\theta_{f_1(x)} \cdot f_1(x_{obs}) + \theta_{f_2(x)} \cdot f_2(x_{obs}) + \dots}}{K} = \frac{e^{\theta_{f_1(x)} \cdot f_1(x_{obs}) + \theta_{f_2(x)} \cdot f_2(x_{obs}) + \dots}}{\sum_{k=1}^{N_{matrices}} e^{\theta_{f_1(x)} \cdot f_1(x_k) + \theta_{f_2(x)} \cdot f_2(x_k) + \dots}}$$

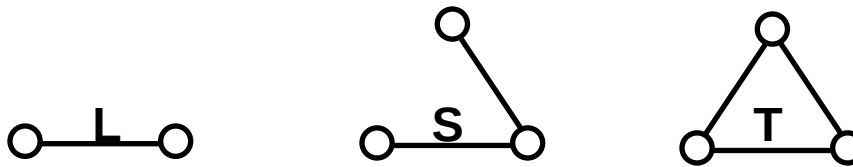
Probability that your observed graph exists

Normalizing constant::
Effect of tie formation mechanisms in any possible graph

- Given the observed characteristics of the graph, what parameters (theta's) for these characteristics will give the highest chance $P(X=x_obs)$ of observing the graph we have observed? (maximizing likelihood)

An example

In practice we often restrict these characteristics of the graph to a limited number of aspects: like the number of ties (L), 2-stars (S) and triangles (T)



The formula then becomes:

$$P(X = x) = \frac{e^{\theta'_L f(x_{obs})}}{K} = \frac{e^{\theta_L L_{obs} + \theta_S S_{obs} + \theta_T T_{obs}}}{\sum_{k=1}^{N_{matrices}} e^{\theta_L L_k + \theta_S S_k + \theta_T T_k}}$$

In this formula:

- L, S and T are known,
- The parameters (θ) for L, S and T need to be estimated

An example

So the question is:

- To what extent does the probability of observing a specific graph dependent on the number of ties, 2-stars or triangles?

$$P(X = x) = \frac{e^{\theta' \cdot f(x_{obs})}}{\mathcal{K}} = \frac{e^{\theta_L \cdot L_{obs} + \theta_S \cdot S_{obs} + \theta_T \cdot T_{obs}}}{\sum_{k=1}^{N_{matrices}} e^{\theta_L \cdot L_k + \theta_S \cdot S_k + \theta_T \cdot T_k}}$$

- Given the observed number of ties, 2-stars and triangles, what parameters theta will give the highest likelihood $P(X=x)$ of observing the graph we have? (maximizing likelihood)
- And so do specific parameter values result in this graph or could it result in just any graph? (significance of the different parameters)

A tiny problem

The idea is nice, but even for computers the **normalizing constant** is difficult to calculate.

Estimating an undirected network with 6 nodes (i.e. 15 dyads) already requires a normalizing constant that compares $2^{15} = 32,768$ possible graphs.

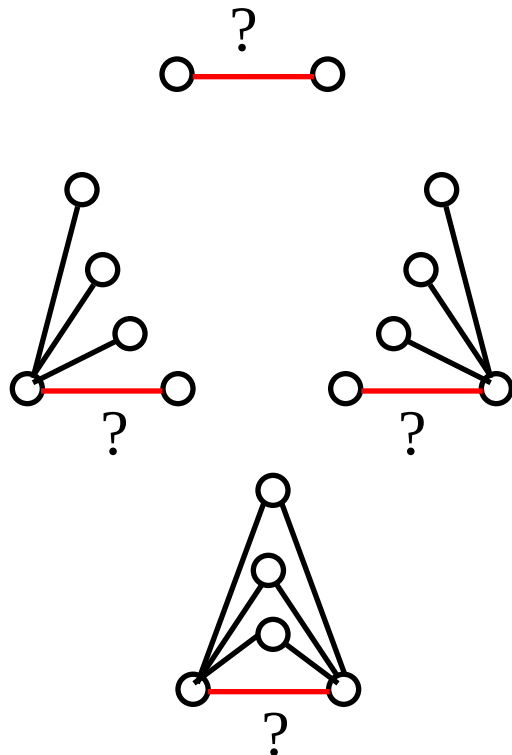
The pseudolikelihood approach

The presence or absence of a tie between i and j (as dependent variable) is set **conditional** on the local aspects surrounding i and j .

Given the rest of the graph, what is the likelihood of $x_{ij} = 1$?

The pseudolikelihood approach

Does the local surrounding predict the presence/absence of a tie between i and j ?

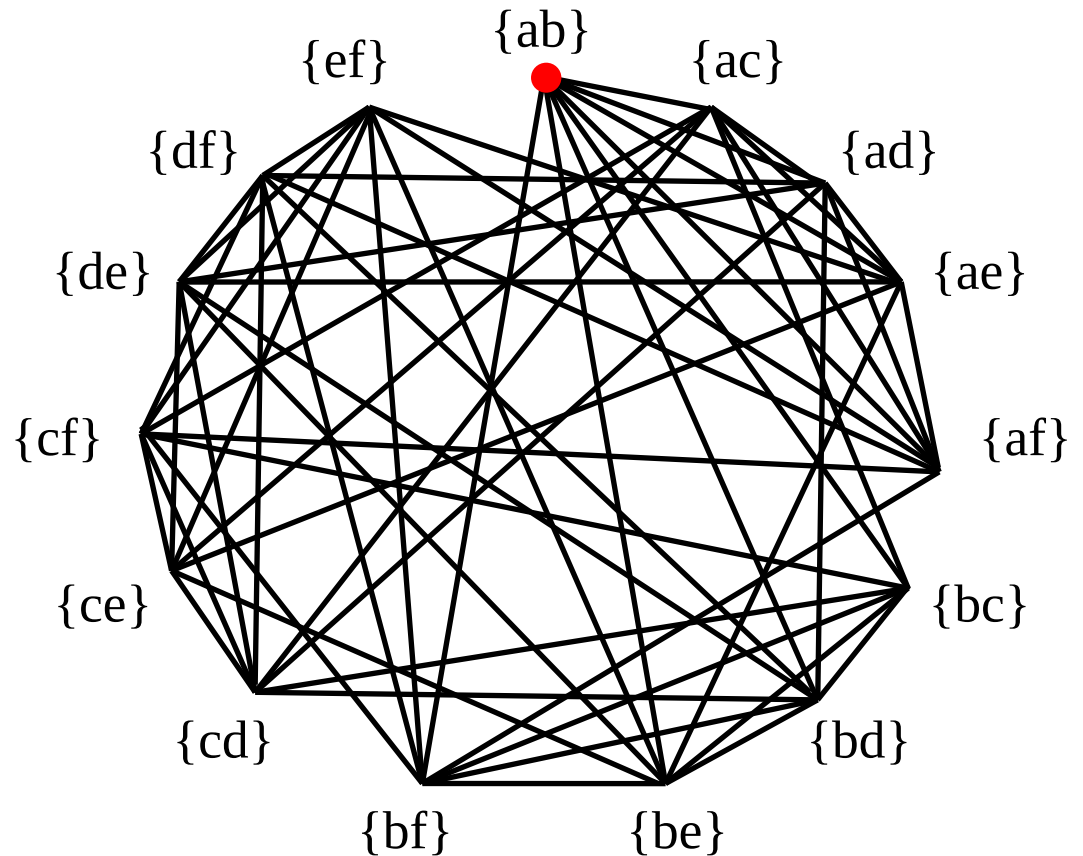
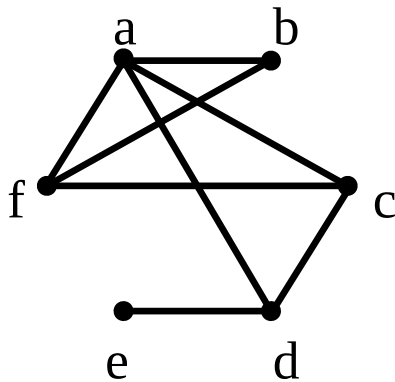


- What is the overall chance of a tie? (density / intercept)
- If actors i and j choose more others, does this increase (or decrease) the chance of a tie between i and j ? = 2-star
- If actors i and j choose the same others, does this increase (or decrease) the chance of a tie between i and j ? = triangle

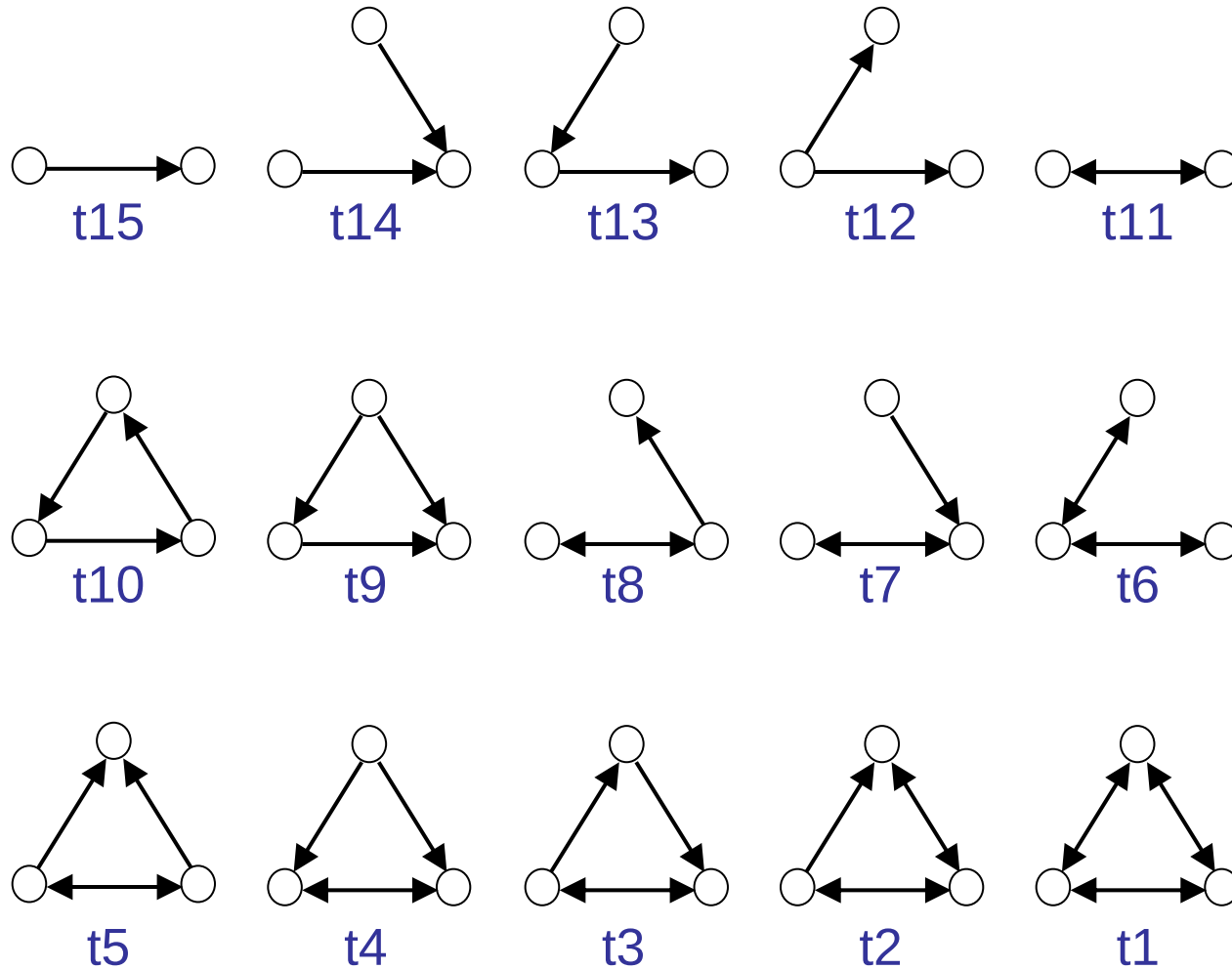
The Markov assumption

- Commonly, ties are assumed to be dependent on each other only if they share a common node.
- What characteristics can be included is based on a **dependence graph**. We can draw a dependence graph, where the nodes are ties between 2 actors of the graph.
- Markov Chain algorithms are used to estimate coefficient values and estimates.

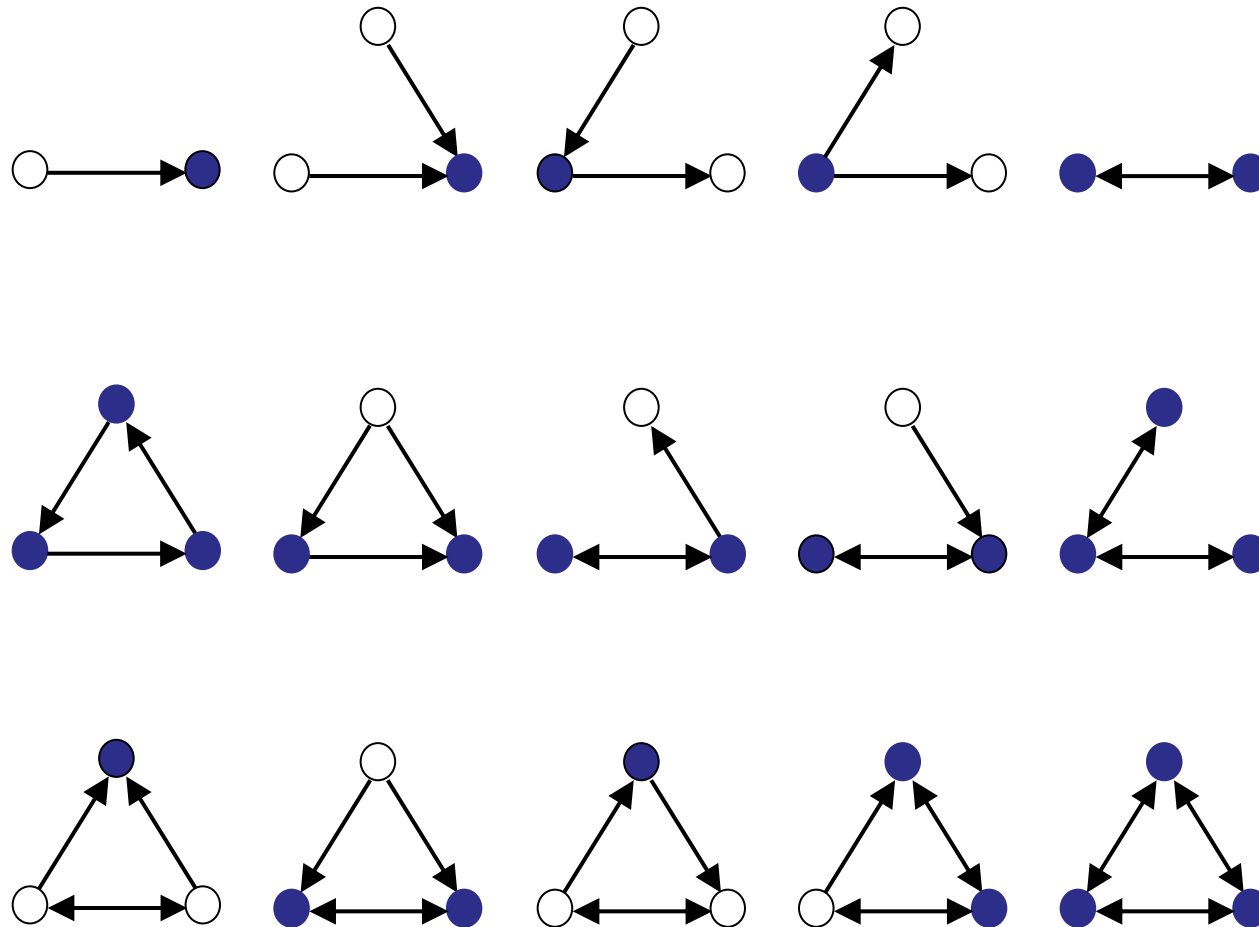
A dependency graph



Extensions to directed graphs



Extensions to models including node attributes



Source: Filip Agneessens

The commands in R

Model with structural effects:

```
yourmodel <- ergm(netw ~ edges + mutual + triangles)
summary(your model)
```

Density effect Reciprocity Closure effect

Model with attribute effects:

```
yourmodel <- ergm(yournetwork ~ nodeicov(var) +
  nodeocov(var) + nodematch(var) + edgecov(var))
summary(your model)
```

Popularity effect on attribute var

Activity effect on attribute var Similarity effect on node attribute var Similarity effect of dyadic attribute var